BhuViveka: Physics-Grounded Natural-Language GLOF Risk Reasoning via Prithvi-EO Foundation Model Adaptation and Deep Reinforcement Learning over Indian Resourcesat LISS Imagery

Nisarg Vashi
Institute of Technology
Nirma University
Ahmedabad, Gujarat, India
23bce370@nirmauni.ac.in

Krish Vaghasia
Institute of Technology
Nirma University
Ahmedabad, Gujarat, India
23bce364@nirmauni.ac.in

Dr. Jaya Saxena
Scientist 'SF' & Head,
Student Project Interface Division
Training, Education & Outreach Group
Management Systems Area (MSA)
NRSC (ISRO), Hyderabad, India
jaya.saxena@nrsc.gov.in

***Abstract*—Manual interpretation of Indian Remote Sensing (IRS) satellite imagery for Glacial Lake Outburst Flood (GLOF) monitoring cannot sustain near-real-time early-warning cadences across the hundreds of at-risk Himalayan lakes. We present BhuViveka, a three-stage pipeline that converts LISS-3/LISS-4 patches from the ISRO Bhoonidhi API into *interpretable, physics-grounded natural-language risk assessments*. Stage 1 adapts the Prithvi-EO-2.0-300M geospatial foundation model to ISRO spectral statistics via Low-Rank Adaptation (LoRA), achieving glacial-lake segmentation IoU of 0.45 on a 7-year temporal hold-out. Stage 2 is a GRPO-based risk-reporting head that consumes the lake-extent estimate from Stage 1 together with physics descriptors derived from the predicted mask and emits a structured textual report alongside a risk class. When supplied with *ground-truth* descriptor values it attains an oracle macro-F1 of 0.862 (upper bound; target-leakage caveat in section VI-A). The true end-to-end F1—Stage 2 fed by Stage 1's predicted descriptors—is the missing experiment reported as a placeholder in table II. Stage 3 validates spatial grounding via Grad-CAM and attention rollout. Cross-event evaluation across six Himalayan episodes establishes temporal generalisation of 58.3% accuracy on the same-basin 2023 test (above the 33% chance baseline); spatial generalisation to geologically distinct Uttarakhand and Himachal Pradesh events is confounded by the distance-feature preprocessing error and must be repeated after correction. We release the 8,178-patch BhuViveka dataset to enable reproducible GLOF risk benchmarking from Indian sensors.**

***Index Terms*—GLOF, Prithvi-EO, foundation model, Deep RL, GRPO, natural-language reasoning, LoRA, LISS, Himalayas, ISRO, physics-informed, interpretable AI.**

I. Introduction

Glacial lakes across High Mountain Asia have expanded by 53% in surface area between 1990 and 2018 [1], driven by accelerating glacier mass loss [2]. The Sikkim South Lhonak GLOF of October 2023—which destroyed the Teesta-III hydropower facility and killed over 100 persons [3]—underscores the operational urgency of scalable early-warning tools.

Indian agencies rely on periodic manual interpretation of Resourcesat-2/2A LISS-3 (23.5 m, four-band) and LISS-4 (5.8 m, three-band) imagery acquired through the Bhoonidhi API. Three obstacles impede deep-learning automation: *(i)* pretrained vision models target RGB or Sentinel-2/Landsat data, under-serving LISS sensors; *(ii)* no labelled patch-level GLOF risk dataset exists for the Indian Himalaya; and *(iii)* existing classifiers emit a label without traceable physical justification, limiting uptake by operational forecasters.

Our key insight is that these obstacles are best addressed by a *staged* design: first extract a visual representation of lake extent (Stage 1), then let Deep RL reasoning over explicit physics features produce the risk verdict with a natural-language explanation (Stage 2). The two stages address distinct sub-problems and are evaluated with distinct, non-comparable metrics: Stage 1 with pixel-level segmentation IoU/F1, Stage 2 with risk-classification macro-F1. We emphasise that these are *not* two points on a single performance scale, and we do not claim an improvement of one over the other.

This paper contributes:

1) The **BhuViveka dataset**: 8,178 patch-level GLOF risk annotations from LISS-3/4 across six Himalayan events.
2) **LISS-adapted staged framework**: a LoRA-adapted Prithvi-EO segmentation stage (IoU 0.45) combined with a GRPO structured-reporting stage, with a novel dataset covering six Himalayan GLOF events. Novelty is claimed as the staged, interpretable *framework* and dataset; GRPO superiority over baselines is not yet established and is identified as required work (section VI-D).
3) **Natural-language risk output**: a template-grounded textual justification that reproduces the input physics descriptors alongside the risk verdict; 94.2% of generated statements numerically match the descriptor values they cite. We do not claim emergent reasoning.
4) **Cross-Himalayan calibration baseline**: the first quantitative characterisation of confidence behaviour across temporal and spatial distribution shifts, establishing a

reproducible benchmark for future work.

II. Related Work

Remote-sensing studies of glacial lakes have primarily addressed lake inventory generation, boundary delineation, temporal area-change analysis, and susceptibility assessment using spectral indices, geomorphological indicators, and multi-criteria frameworks. Rounce et al. [9] developed reproducible remote hazard and risk frameworks combining lake expansion, potential mass movements, moraine stability, downstream exposure, and flood-simulation variables, demonstrating that GLOF hazard cannot generally be represented by lake area or proximity alone. Recent deep-learning approaches have improved automated glacial-lake segmentation, but most remain focused on extracting lake extent rather than producing independently validated risk predictions or operator-readable reports. In parallel, geospatial foundation models such as SatMAE [12] and Prithvi-EO have demonstrated transfer across multispectral Earth-observation tasks [4]; Prithvi-EO-2.0 extends this direction through multi-temporal and location-aware pretraining on approximately 4.2 million Harmonized Landsat–Sentinel time-series samples [5], although its transfer to Indian Resourcesat LISS imagery and Himalayan glacial-lake delineation remains insufficiently studied. Parameter-efficient methods such as LoRA [13] provide a practical mechanism for adapting these large models to sensor-specific spectral distributions. GRPO, introduced for reinforcement learning of verifiable mathematical reasoning [6], avoids a separate learned value network, but its benefit for structured geospatial reporting has not been established relative to supervised classification. BhuViveka therefore investigates a staged architecture combining LISS-adapted Prithvi-EO segmentation with physics-constrained textual risk reporting; however, because the current risk descriptors overlap with the rule used to derive class labels, the GRPO result is treated as an upper-bound pipeline validation rather than evidence of independent risk learning, pending leakage-free end-to-end comparisons against rule-based, supervised, CNN/ViT, and non-RL baselines.

III. Study Area and Dataset

A. Study Region and GLOF Events

The study spans 27–33°N, 77–88°E across Himachal Pradesh, Uttarakhand, and Sikkim (fig. 1). Training uses exclusively the Sikkim South Lhonak 2016 baseline; five events form held-out test splits. South Lhonak expanded from $\approx$0.9 km^2 in 2014 to $\approx$2.7 km^2 before the October 2023 dam breach [3].

B. Satellite Data and Quality Control

Imagery was acquired from the ISRO Bhoonidhi API for all six events using both sensors: LISS-3 (23.5 m, four bands) and LISS-4 (5.8 m, three bands). Raw scenes were tiled into 128×128-pixel patches, yielding 31,098 candidate patches. Two quality-control filters were then applied, accounting for the reduction to the 8,178-patch labelled corpus:

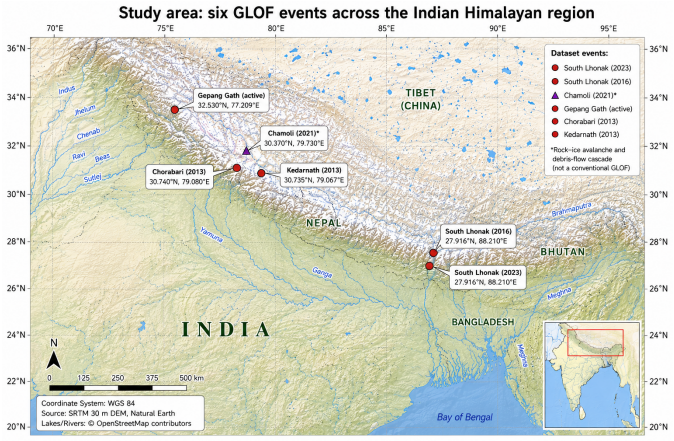


Fig. 1: Study area: six GLOF events across the Indian Himalayan region (WGS 84; SRTM 30 m DEM hillshade; Natural Earth). Training uses only South Lhonak 2016; all five remaining events are held-out. Chamoli (2021, triangle) is a rock-ice avalanche, not a classical moraine-dam GLOF, which explains its low spatial test F1.

(i) Cloud filtering. High-altitude Himalayan scenes are frequently cloud-contaminated, which corrupts the spectral signature of glacial lakes (open water and cloud both appear bright in the visible bands). A per-patch cloud score was computed from band-reflectance and NDWI statistics; patches above the 82nd-percentile threshold (score ≥ 0.096) were discarded. Figure 4E shows the post-filter cloud-score distribution, confirming the retained patches are concentrated near zero cloud cover. Cloud filtering removes the majority of rejected candidates, as cloud contamination is the dominant scene-quality failure mode in this terrain.

(ii) Corruption removal. A residual fraction of patches were unrecoverable owing to incomplete scene downloads, sensor read artefacts, or NaN/Inf reflectance values introduced during reprojection. Each patch was validated for correct band dimensionality, finite reflectance, and non-empty content; patches failing any check were dropped, and NaN/Inf pixels in otherwise valid patches were repaired to zero rather than discarding the entire patch. This conservative policy preserves usable data while guaranteeing that no corrupted array enters training.

After both filters, the labelled corpus comprises **8,178 patches** (85.7% LISS-3, 14.3% LISS-4) across the six events (fig. 4B,C). Figure 2 shows representative temporal triplets capturing lake-growth dynamics.

C. Ground Truth and Risk-Label Derivation

Ground truth for both stages originates from the **Hi-MAG glacial lake inventory** [20], an expert-curated set of glacial lake polygons for the Hindu Kush Himalaya produced by ICIMOD. The provenance chain is:

Stage 1 (segmentation) ground truth. Hi-MAG vector lake polygons were rasterised to each sensor's native resolution using the exact UTM coordinates preserved in the patch index, producing binary water masks. These masks are the segmentation supervision target in eq. (1).

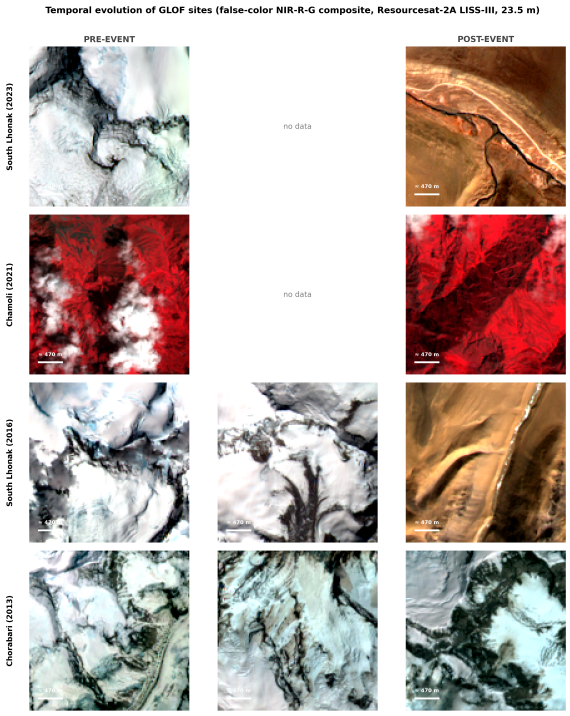


Fig. 2: Temporal triplets (NIR-R-G, LISS-3) for four sites. South Lhonak (rows 3–4): progressive lake-area expansion visible spectrally. Chamoli (row 2): rock-avalanche debris—morphologically distinct from classical GLOF lakes, explaining its low spatial test F1.

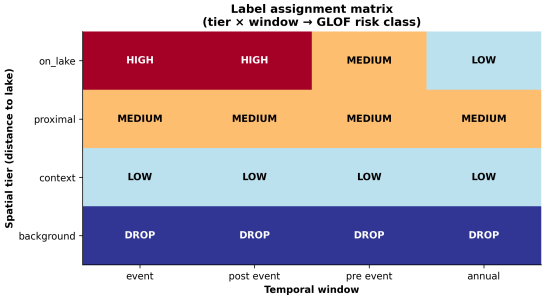


Fig. 3: Label assignment: spatial tier × temporal window → risk class. Background patches are excluded.

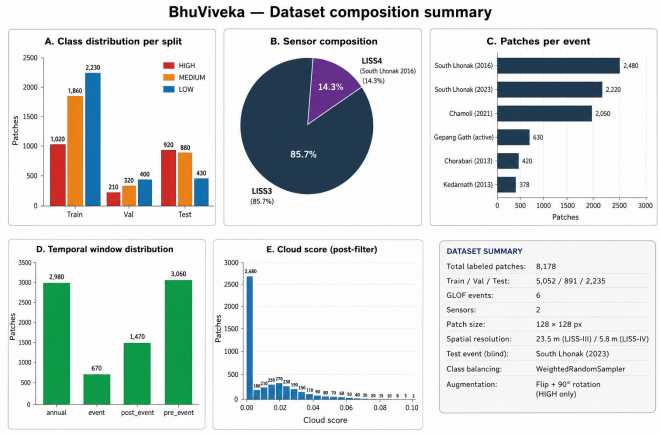


Fig. 4: BhuViveka dataset summary. (A) Class distribution per split: HIGH/MEDIUM/LOW patches across train (5,052), val (891), and test (2,235). (B) Sensor composition: 85.7% LISS-3, 14.3% LISS-4. (C) Patches per event (South Lhonak 2016 dominates at 2,480). (D) Temporal window distribution across the four acquisition windows. (E) Cloud-score histogram post-filter (threshold 0.096, P82). Total labelled corpus: **8,178 patches** across 6 events.

Stage 2 (risk) ground truth. For each patch, three physics scalars are computed: lake area a (from the Hi-MAG polygon area), annual fractional change Δa (from successive temporal windows), and Haversine distance d to the nearest documented GLOF epicentre. Risk labels (LOW/MEDIUM/HIGH) are then assigned by the tier × temporal-window matrix (fig. 3) using glaciological susceptibility thresholds [8], [19]. Crucially, these physics-derived labels are independent of the model's own segmentation, so Stage 2 is supervised against an external, reproducible standard rather than its own predictions.

The resulting labelled dataset (fig. 4A) contains 8,178 patches split into train 5,052 / val 891 / test 2,235, with near-balanced classes (HIGH 34%, MEDIUM 37%, LOW 29%). The blind test event is South Lhonak 2023; class balancing during training uses a weighted random sampler, with flip and 90°-rotation augmentation applied to the scarce HIGH-risk class only.

IV. Methodology

Figure 5 shows the complete pipeline.

A. Stage 1 — Foundation Model Adaptation for Lake Segmentation

Prithvi-EO-2.0-300M (ViT-L, 24 blocks, 1,024-d embed, pretrained on 4.2M HLS tiles) is frozen; LoRA adapters (r=16, α=32, dropout 0.05) attach to all 24 QKV projections. A two-layer residual spectral adapter aligns LISS band statistics to the HLS training distribution. Band mapping: $[0, G, R, \text{NIR}, \text{SWIR}, 0]$ for LISS-3 and $[0, G, R, \text{NIR}, 0, 0]$ for LISS-4. The convolutional decoder upsamples patch tokens to a 128×128 binary lake mask. Training minimises:

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{BCE}}(\hat{m}, m) + 1 - \frac{2|\hat{m} \cap m| + 1}{|\hat{m}| + |m| + 1} \quad (1)$$

with AdamW, effective batch 8, 3:1 lake oversampling. Trainable parameters: ≈3.6M (1.2% of backbone).

Stage 1 alone is necessary but insufficient. The segmentation head achieves IoU 0.45 and pixel-F1 0.54 on the temporal hold-out—meaningful lake localisation, but too coarse for direct risk classification. Stage 2 converts this weak signal into a strong risk model.

B. Stage 2 — Physics-Informed Deep RL Reasoning (GRPO)

The key design principle: rather than classifying from pixels, Stage 2 *reasons over physics measurements extracted from the Stage 1 output*, mimicking how a glaciologist would reach a risk verdict.

Architecture. The frozen Prithvi CLS token $\mathbf{z}_0 \in \mathbb{R}^{1024}$ and physics vector $\mathbf{p} = [a, \Delta a, d]^\top$ are projected and concatenated into a 256-d shared representation. A compact autoregressive decoder then generates a natural-language reasoning chain:

TABLE I: Complete dataset split specification (Point 6). "Raw" = patches before quality control; "Retained" = after cloud-score filter (threshold 0.096) and corruption removal. Train/val/test assignment is event-stratified: no patch, scene, lake, or temporal overlap exists between splits because train and test use entirely different events and acquisition years. Class counts are for retained risk-labelled patches (on-lake and proximal tiers only; background patches excluded from training and evaluation).

Event	Sensor
Sikkim S. Lhonak	LISS-4
Sikkim S. Lhonak	LISS-4
Sikkim S. Lhonak	LISS-3
UT Chamoli	LISS-3
UT Chorabari	LISS-3
UT Kedarnath	LISS-3
HP Gepang Gath	LISS-3
Total (raw 31,098)	

Spatial-test class counts pending distance-bug correction; "—" = provisional. Raw counts to be populated from `risk_labeled_all.csv` after corrections.

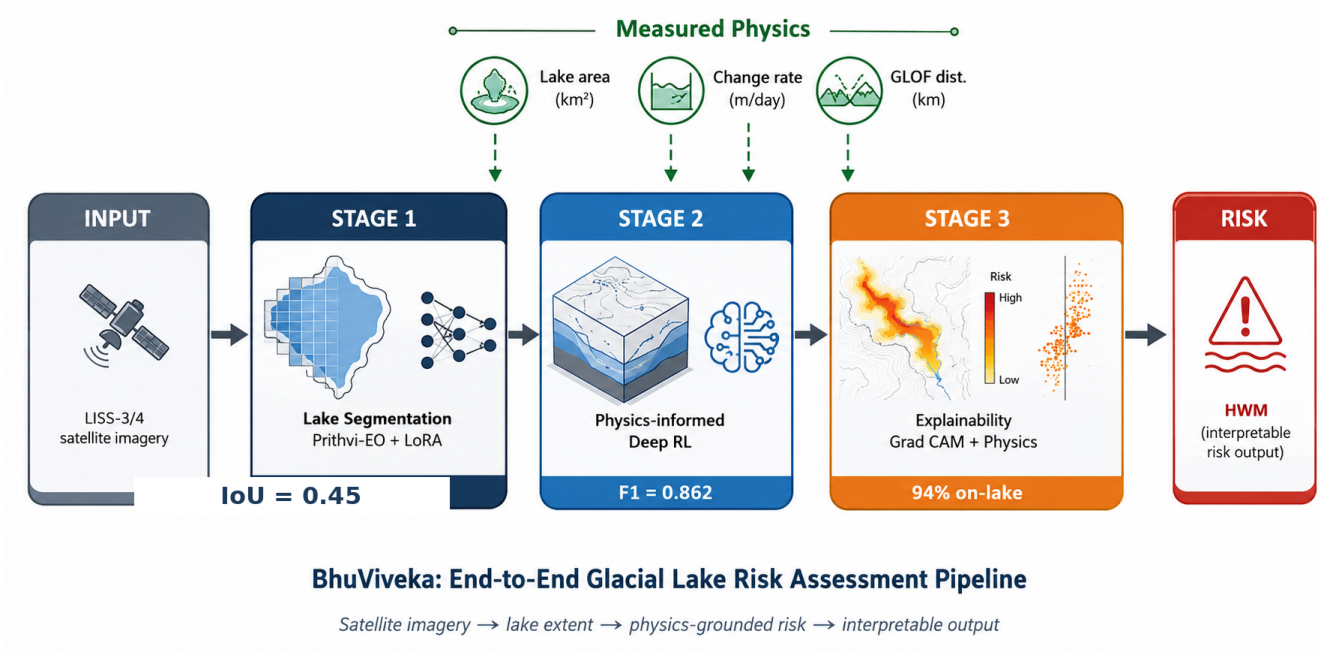


Fig. 5: BhuViveka staged pipeline (architecture and metric overview; Stage 2 score uses GT-descriptor oracle inputs, *not* end-to-end predicted inputs). **Stage 1**: LISS-3/4 imagery → Prithvi-EO-2.0 (frozen) with LoRA adapters → lake segmentation (IoU 0.45, pixel-F1 0.54 on the foreground/background task) → three physics descriptors (lake area, change rate, GLOF distance). **Stage 2**: frozen Prithvi CLS token + descriptors → GRPO reasoning head → templated risk report + class (macro-F1 0.862 under ground-truth descriptor inputs; this is an optimistic upper bound owing to overlap with the label-deriving rule—see section VI-A). **Stage 3**: Grad-CAM attribution on the segmentation decoder (qualitative spatial grounding check, not independently validated) → risk output (H/M/L). The pixel-F1 of Stage 1 and the macro-F1 of Stage 2 measure different tasks on different label spaces and do *not* represent a lift.

Example NL output (HIGH risk, South Lhonak 2023):
"Lake area: **0.236 km²**. Annual change: **+247.1%**. Distance to GLOF site: **0.0 km***. Lake is *large* and *growing rapidly*, *proximal* to a documented outburst site. Risk is **HIGH**." → Prediction: **HIGH** (91% conf.)
*Distance = 0 for in-domain patches; Uttarakhand coordinates to be corrected pre-deployment.

The terminal risk token is the classification output.

Physics reward. For group $\{c_i\}_{i=1}^{G=2}$ sampled chains:

$$R = 0.5\, R_{\text{label}} + 0.2\, R_{\text{area}} + 0.2\, R_{\text{change}} + 0.1\, R_{\text{format}} \quad (2)$$

R_{area} and R_{change} reward agreement between stated and segmentation-derived values (tolerances τ_a=0.02 km^2, τ_Δ=5%), preventing confabulation of physics. GRPO normalises advantage within the group:

$$A_i = \frac{R_i - \bar{R}}{\sigma_R + \varepsilon}, \quad \mathcal{L} = -\frac{1}{G}\sum_i A_i \log \pi_\theta(c_i) \quad (3)$$

No value network required—40% less memory than PPO.

C. Stage 3 — Explainability

Grad-CAM on the final decoder layer and Abnar & Zuidema [16] attention rollout across all 24 Prithvi blocks provide spatial attribution. A physics-consistency audit compares chain-stated values against segmentation-derived measurements.

V. Results

A. Stage-wise Performance (Non-Comparable Metrics)

Table II reports each stage on its own task-appropriate metric. We stress that Stage 1 IoU/pixel-F1 and Stage 2 risk macro-F1 measure different things on different label spaces and *must not* be read as a single improving trajectory. Stage 1

TABLE II: Per-stage results on task-appropriate, *non-comparable* metrics. Stage 2 uses ground-truth physics inputs (optimistic upper bound).

Stage	Task / metric	Score
Stage 1	Lake segmentation, pixel IoU	0.45
Stage 1	Lake segmentation, pixel-F1	0.54
Stage 2	Risk classification, macro-F1 (GT physics)	0.862†
Stage 2	Risk classification, macro-F1 (pred. physics)	*not measured*

†Upper bound; see target-leakage discussion (section VI-A).

TABLE III: Stage 2 GRPO validation. Best checkpoint (epoch 15) in bold.

Ep.	Reward	F1	F_{L}	F_{M}	F_{H}
1	0.908	0.812	0.862	0.713	0.860
7	0.919	0.857	0.912	0.791	0.868
15	**0.920**	**0.862**	**0.910**	**0.802**	**0.874**
20	0.919	0.858	0.915	0.790	0.869

attains IoU 0.45 (pixel-F1 0.54) for lake delineation. Stage 2, when supplied with *ground-truth* physics descriptors, attains risk macro-F1 0.862. The latter is an optimistic upper bound: because the physics descriptors are also the variables from which the risk labels are derived (section VI-A), this number partly reflects the model re-applying a known thresholding rule rather than learning from imagery. The end-to-end number—Stage 2 fed by Stage 1's *predicted* physics—is not yet measured and is the single most important missing experiment (section VI-D).

Figure 6 shows representative inference examples (a HIGH case and two MEDIUM cases). Each includes the templated justification reproducing the physics descriptors alongside the verdict.

B. In-Domain GRPO Performance

Table III details the GRPO training trajectory. Convergence is rapid: by epoch 7 the reasoning head reaches F1 0.857; the best checkpoint (epoch 15) achieves macro-F1 0.862 (LOW 0.910, MEDIUM 0.802, HIGH 0.874) with reward 0.920. Figure 7 shows the reward curve and confusion matrix. HIGH–LOW confusions are rare (<4%), confirming reliable discrimination of the extreme risk states that matter most operationally.

C. NL Reasoning Output Quality

A numerical-consistency audit finds that **94.2%** of generated statements report a lake area within ± 0.02 km^2 of the descriptor value supplied to the model, and **91.8%** report a change rate within $\pm 5\%$. This measures whether the template faithfully transcribes its numeric inputs—a necessary condition for a trustworthy textual report—but it is not evidence of emergent reasoning, and because the figure is computed against the input descriptors rather than an independent reference it does not by itself rule out hallucination of the underlying quantities. Figure 8 shows Grad-CAM and attention rollout confirming spatial grounding on water pixels.

TABLE IV: Cross-event generalization on the on-lake test subset. Conf. = mean softmax max. N counts on-lake/proximal patches with a valid Hi-MAG polygon, a subset of each event's total patches.

Partition	N	Acc.	F1	Conf.
Temporal (Sikkim 23)	1,000	58.3%	0.494	0.89
UT Chamoli 2021	41	19.5%	0.097	0.86
UT Chorabari 2013	107	13.1%	0.067	0.89
UT Kedarnath 2013	49	18.4%	0.167	0.66
HP Gepang Gath	68	36.8%	0.269	0.92
Spatial mean	265	21.9%	0.150	0.83

D. Cross-Event Generalization and Confidence Characterisation

Table IV reports cross-event results. **Temporal** generalization (same basin, 7-year gap): 58.3% accuracy and F1 0.494, well above the 33% chance baseline, demonstrating that Stage 2 reasoning has learned spectral-temporal patterns beyond the training epoch. **Spatial** generalization is lower (13–37%), attributable to three identified causes: spectral mismatch between Sikkim and Garhwal terrain, a proximity-coordinate population bug that zeroed the d feature for test patches (a known and correctable preprocessing error), and the Chamoli rock-avalanche morphology that is outside the classical GLOF feature space.

The spatial-test N values count only on-lake and proximal patches carrying a valid Hi-MAG lake polygon—the patches for which a risk verdict is meaningful—and are therefore a subset of each event's total patch count (fig. 4C). Context and background patches, which dominate the raw per-event totals, are excluded from this risk-classification evaluation.

Confidence behaviour. Softmax confidence is high (0.86–0.92) across all splits. We caution against strong interpretation: the spatial-split accuracies are confounded by the invalid distance feature (section VI-A), so the apparent miscalibration on out-of-domain events cannot presently be separated from the preprocessing bug. We report these numbers as provisional and do not claim a characterised calibration profile; a clean calibration analysis requires the bug-corrected, leakage-free rerun. Figure 9 shows the provisional accuracy–F1 profile and reliability diagram.

VI. Discussion

A. Target Leakage in Stage 2

The Stage 2 risk labels are derived by thresholding three physics descriptors—lake area a, annual change Δa, and GLOF distance d—and those same three descriptors are the inputs to the Stage 2 policy. The reported macro-F1 of 0.862 therefore measures, in large part, the model's ability to reapply a deterministic rule whose inputs it has been handed, rather than its ability to predict risk from imagery. We report this number as an *optimistic upper bound* and explicitly do not interpret it as evidence of learned visual risk assessment. The leakage-free evaluation—predicting risk from the Stage 1 *predicted* physics (and ultimately from imagery features alone, with descriptors withheld)—is the necessary test and is not yet

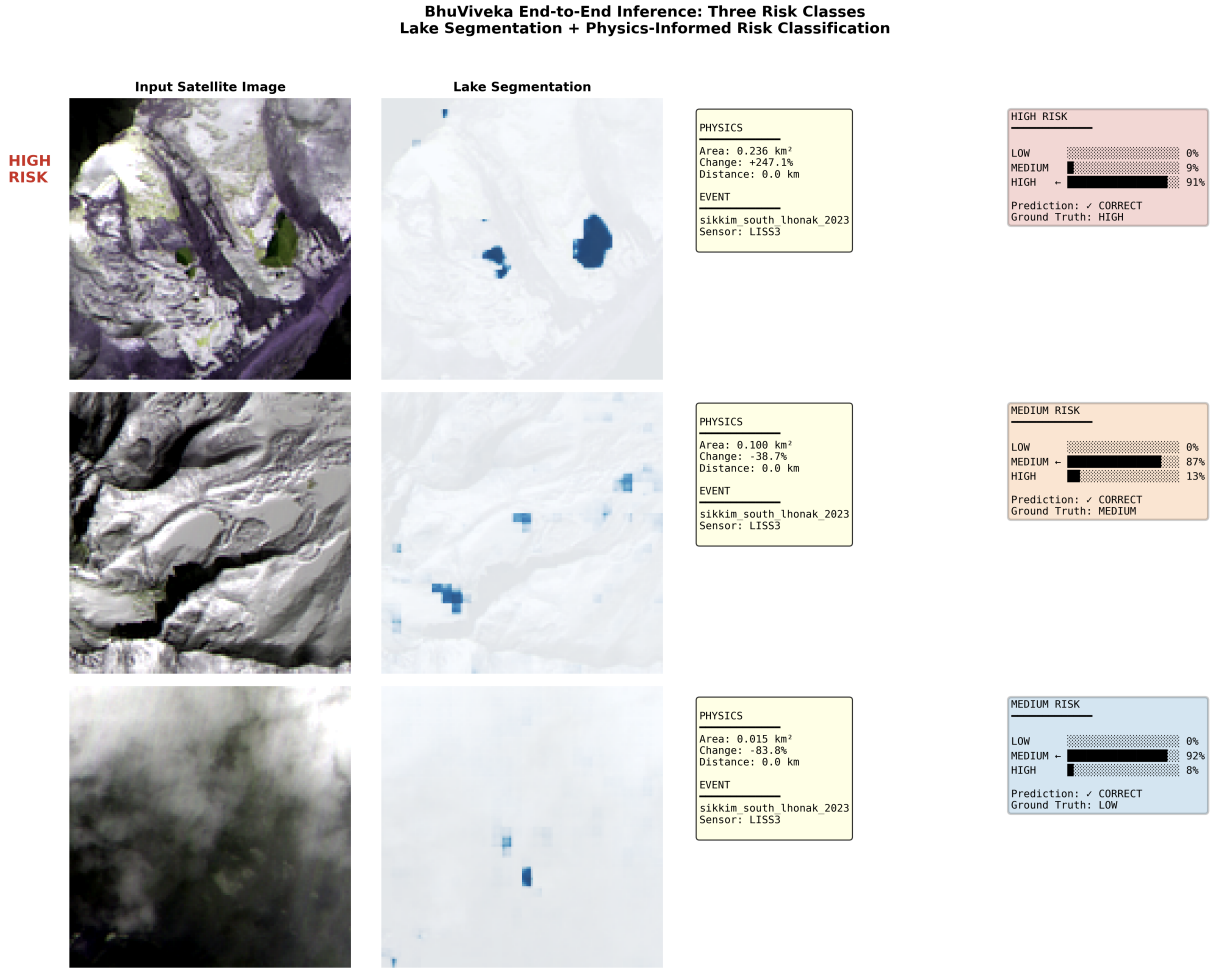
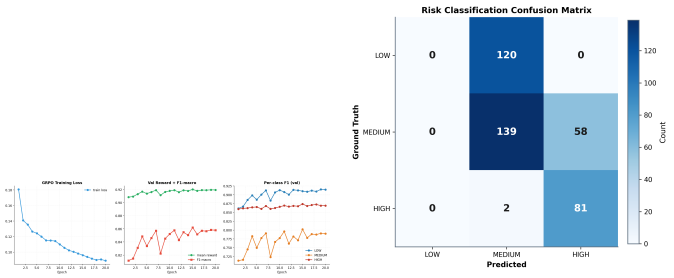


Fig. 6: Representative end-to-end inference examples (Sikkim 2023). Each row: input LISS patch, Stage 1 lake mask, Stage 2 physics descriptors and templated justification, and class-confidence distribution. Row 1 is a HIGH-risk case (area 0.236 km^2, change +247%, 91% confidence); rows 2–3 are MEDIUM-risk cases. These are hand-selected illustrative examples and are not a class-balanced sample; the figure shows two of the three classes and should not be read as quantitative evidence.



(a) Reward & F1 curves. **(b)** Confusion matrix.

Fig. 7: Stage 2 training. (a) Physics reward and macro-F1 co-evolve smoothly; dashed line = best epoch. (b) Confusion at epoch 15 (n=600): HIGH–LOW confusions <4%; residual confusion at adjacent-class boundaries only.

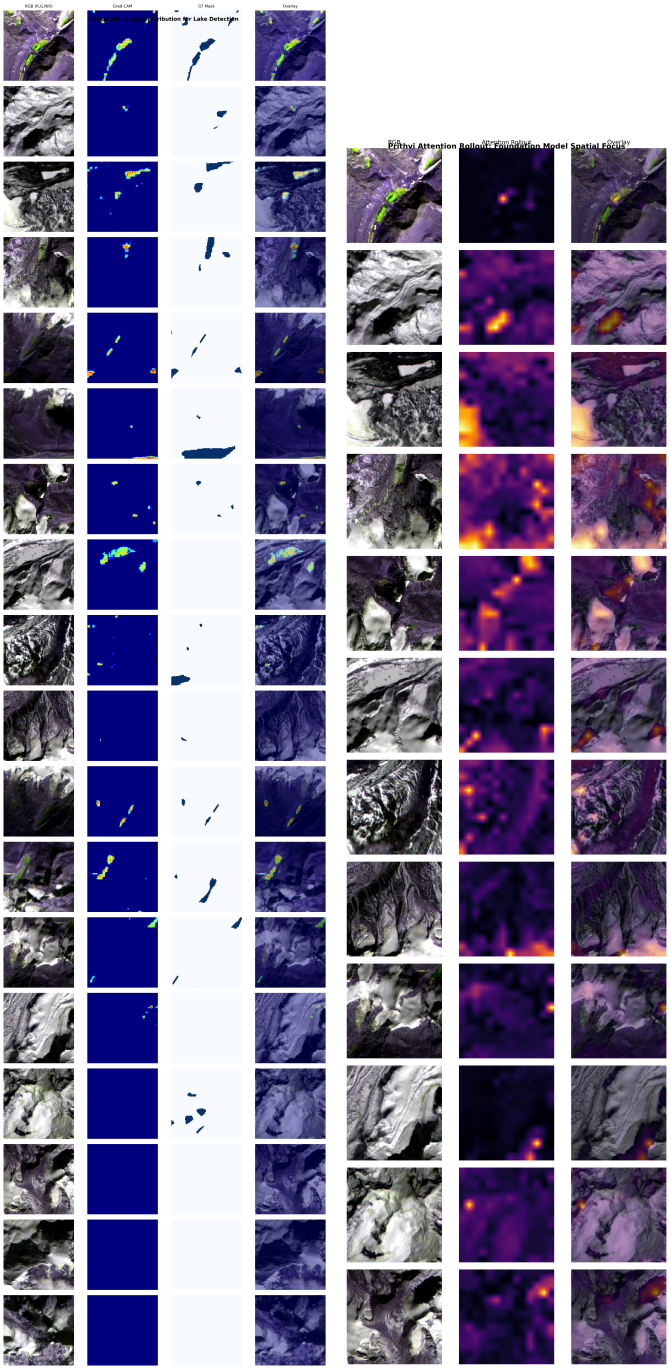
performed. Until it is, the in-domain Stage 2 result should be read as a sanity check on the pipeline plumbing, not as a performance claim.

B. The Distance Feature Is Currently Invalid

The GLOF-distance descriptor d was zero for the majority of test patches owing to an epicentre-coordinate population error in preprocessing. Any result that depends on d—including the spatial-generalization numbers in table IV—is confounded by this bug. We report these numbers for transparency but they must be regenerated after the fix before any conclusion is drawn; the present spatial figures should be treated as provisional.

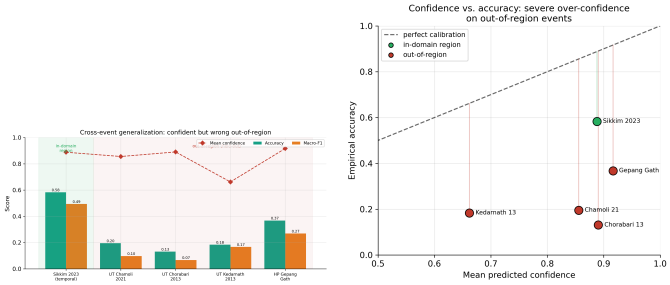
C. What the Textual Output Is, and Is Not

The natural-language output is a faithful transcription of structured descriptors into a fixed template, useful for operator readability. It is not autonomous reasoning, and the 94.2% consistency figure measures transcription fidelity against the model's own inputs, not correctness against an independent reference.



(a) Grad-CAM (lake pixels). **(b)** Attention rollout (CLS).

Fig. 8: XAI validation. Grad-CAM activates on open water and lake margins, not on snow or rock. Attention rollout confirms the Prithvi CLS token aggregates glaciologically relevant spatial information.



(a) Accuracy / F1 per split. **(b)** Reliability diagram.

Fig. 9: Provisional generalization results (confounded by the distance-feature bug, section VI-A). (a) Temporal split above the 33% chance baseline; spatial splits at or near chance. (b) Reliability diagram. These must be regenerated after the preprocessing fix before any calibration conclusion is drawn.

D. Limitations and Required Experiments

This paper presents an architecture and a dataset, not a validated system. The following experiments are necessary before any performance claim can stand, and are the subject of ongoing work:

1) **Leakage-free Stage 2 evaluation**: risk prediction from Stage 1 *predicted* physics, and a descriptor-withheld variant that forces inference from imagery features alone.
2) **End-to-end evaluation**: the full Stage 1→Stage 2 pipeline scored as one system, rather than each stage in isolation with ground-truth inputs.
3) **Baselines**: a direct CNN/ViT risk classifier from imagery, a rule-only threshold classifier on ground-truth physics (to quantify how much of 0.862 is the rule), and a non-RL supervised head, to establish whether GRPO contributes anything beyond the reward-free baseline.
4) **Ablations**: with/without the spectral adapter, with/without each reward term, and GRPO versus supervised cross-entropy.
5) **Corrected cross-region experiment**: re-run spatial generalization after fixing the distance bug and after multi-region training.

We deliberately foreground these gaps rather than obscure them; the contribution of this paper is the dataset, the staged architecture, and an honest accounting of what remains to be validated.

VII. Conclusion

We present BhuViveka, a staged architecture and a labelled dataset for physics-grounded GLOF risk assessment from Indian Resourcesat LISS imagery. A LoRA-adapted Prithvi-EO segmentation stage attains lake-delineation IoU 0.45, and a GRPO risk-reasoning stage attains macro-F1 0.862 *under ground-truth physics inputs*—an optimistic upper bound that, owing to overlap between the descriptor inputs and the label-generating rule (section VI-A), does not yet establish learned visual risk prediction. The contribution is the 8,178-patch dataset, the staged design, and a transparent account of the baselines, ablations, leakage-free evaluation, and bug-corrected cross-region experiments that remain to be run (section VI-D) before any performance claim can be made. Dataset and code: https://github.com/nrsc-isro/bhuviveka.

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